

Mission Control Interview – Operational Protocols & Communication

Overview

This document captures firsthand insights into mission control operations, communication protocols, and decision-making practices. The focus is on realism, hierarchy, and how protocol functions as a tool rather than a rigid rule set.

Predictability of Issues in Mission Control

Issues encountered in mission control are **highly role-dependent** and **scale with severity and responsibility**. Many officers only engage with problems relevant to their specific console or discipline. Some issues are recurrent and well-documented in internal “common problems” lists. Problems range from minor telemetry anomalies to critical life-safety events. Although rare, **fire is considered the highest-risk scenario**. Other common issues include: Gyroscope degradation or failure, communication message dropouts, sensor false positives. Officers on the ground **cannot physically see the situation**, relying entirely on data streams and crew reports. Maintaining situational awareness across many concurrent systems is a major cognitive challenge. Protocols are useful for both large and small issues, as smaller anomalies may be indicators of larger system failures.

Historical Examples of Mission Control Problems

Several real operational incidents were discussed:

- **Ammonia Leak Incident**
 - Triggered by a sensor falsely detecting ammonia
 - Required rapid evaluation of environmental risk
- **Environmental Control Failures**
 - Fire, atmospheric composition changes, and toxic leaks are treated as top-priority scenarios
- **Waste System Leak**
 - A leak in the ISS waste system caused significant crew disruption

- Triggered immediate ground-to-crew communication
- **Aging Technology Constraints**
 - Much of the ISS hardware dates back to the early 2000s
 - Troubleshooting often requires isolating entire modules
 - Example response: sealing off a wing and consolidating the crew in Russian quarters

Problem-solving often involves isolating systems, coordinating across international partners, and working with incomplete or misleading sensor data.

Tone and Hierarchy of Communication

Communication within mission control is professional, direct, respectful, and structured by hierarchy.

Key characteristics:

- The **Flight Director (FD)** is the most senior authority.
- Decisions flow from the top, but **disagreement is expected and required**.
- Officers are responsible for immediately voicing concerns or dissent.
- Peer communication (backroom to backroom) is more casual.
- Teams change frequently as personnel rotate and advance.
- Cohesion is built despite frequent team restructuring.

Communication Frequency and Shift Dynamics

- Communication between rooms is frequent, even during nominal operations.
- During quiet periods, communication may be minimal by design.
- The Flight Director can seamlessly reach any console as needed.
- Certain roles provide **24/7 monitoring** and perform detailed handoffs between shifts.
- Backroom teams often detect anomalies before front-room consoles.
- Activity levels vary based on astronaut sleep/wake cycles.
- Nonessential roles may be on call during low-activity periods.
- Example: a toilet system failure triggering a 2 a.m. call-in.

Protocols and Rules in Mission Control

Protocols are **not rigid rules**, but **tools for problem-solving**.

Key insights:

- Protocol strictness increases with seniority and experience.
- Protocols do not cover every scenario.
- Officers must know **when to follow protocol and when to deviate**.
- The ISS operates largely as an autonomous system during nominal phases.
- After initial system setup, mission control is often in a monitoring role.
- Cheat sheets and reference materials are commonly displayed at consoles.

Protocols exist to support judgment, not replace it.

Physical Movement and Escalation

- Most mission control work is digital and performed from consoles.
- Physical movement is rare and usually associated with escalation.
- If immediate attention is needed, officers may:
 - Raise their voice
 - Make an announcement to the room
- These situations typically indicate:
 - High uncertainty
 - Large-scale casualty scenarios
 - Breakdown in shared situational awareness

Such escalations are treated with high precaution.

Room Occupancy and Staffing

- Mission control can move to a **skeleton crew** rapidly.
- Core focus is always on “what is keeping the ISS alive.”
- The Flight Director is not always present during low-risk periods.
- On many night shifts, the FD remains present due to reduced staffing.
- Officers may be required to operate with minimal direct guidance.

Main Display Systems

Large front-room displays serve as:

- Situational awareness tools
- Timeline references

- Live astronaut monitoring feeds

Key characteristics:

- Displays often show:
 - Current orbit
 - Operational timeline
 - Live video
- Operations are scheduled minute-by-minute during active periods.
- Timelines shift dynamically as tasks evolve.
- Updating display configurations may require third-party IT support.
- Officers typically use dual computers for redundancy.
- Despite dated systems, safety and reliability take priority over modernization.
- Innovation is possible but must overcome bureaucratic and safety hurdles.
- New problems continue to emerge even after 25 years of ISS operations.

Console Layout and Proximity

- Console placement historically reflected communication needs.
- Modern communication systems reduce dependency on physical proximity.
- Distance still matters for informal, rapid interaction.

Project-Related Insights for AI Personas

Critical Personas

- ADCO is the highest priority persona.
- Expansion to front-room, back-room, and astronaut roles is ideal.
- Personas should translate role-specific needs into actionable information.

Narrative and Interaction Style

- Preferred interaction style is **one-on-one**, similar to a coworker.
- AI should feel like a team member, not a lecture system.
- Spatial presence in VR enhances realism but communication is primary.

Constraints and Priorities

- Realism is the top priority.
- Communication accuracy matters more than visual fidelity.
- Empathy and human factors are critical.

Leadership Value

Leadership would prioritize:

1. Clear problem framing
2. Interactive 3D communication
3. Empathy and human understanding

Resources and Expansion

- Persona cards from internal organizations could be valuable inputs.
- Prior conversational AI + VR projects offer strong reference material.
- The prototype is a proof of concept and may incorporate public or synthesized data.

Target Scope

- Initial focus: ADCO persona
- Future expansion: front room, back room, astronauts
- Breadth depends on access to subject matter experts